AIRBNB MARKET INTELLIGENCE PLATFORM

Repeatable Data Pipeline, Exploratory Segmentation, and Guided Analytics Dashboard

Expernetic Data Engineer Talent Assessment Program

London and New York Airbnb Data

Submission Report | June 2026

Contents

Report structure and key deliverables

How to read this report

The report follows the data lifecycle from raw source files through modular pipeline processing, curated outputs, exploratory analytics, segmentation, and the Streamlit application.

1

Executive Summary

A practical end-to-end data engineering and market intelligence solution

This project delivers an end-to-end Airbnb market intelligence platform using listing, review, and calendar data from London and New York. The solution converts raw multi-table data into a processed master dataset, creates curated analytical outputs, performs DuckDB SQL analysis and exploratory KMeans segmentation, and presents the final results through a Streamlit dashboard.

The implementation was designed around a clear separation of responsibilities. The modular Python pipeline handles ingestion, validation, cleaning, review aggregation, calendar aggregation, and master-dataset construction. The notebooks provide transparent analytical evidence. The dashboard consumes processed and curated outputs instead of performing expensive raw-data transformations interactively.

Outcome	Result
Cities analysed	London and New York
Combined master dataset	131,907 listings
Core raw data domains	Listings, reviews, and calendar records
Analytical SQL engine	DuckDB
Exploratory ML method	KMeans clustering with four final segments
Dashboard components	Market Overview, Guided Analytics, Segment Explorer, Pipeline Status

Key interpretation rule

Calendar unavailable dates are used as an occupancy/unavailability proxy. They are not confirmed bookings because unavailable dates may also be host-blocked or restricted for other reasons.

2

Objectives, Scope, and Delivery Prioritisation

The project prioritised a dependable analytical workflow over unsupported production claims.

2.1 Business Objective

The objective was to design an Airbnb market intelligence solution that supports exploratory analysis of supply, listing characteristics, calendar availability patterns, review activity, and location-level price patterns. The platform was designed to answer structured stakeholder questions while keeping the limitations of the available source data explicit.

2.2 In-Scope Deliverables

- Multi-city source-data familiarisation and schema comparison.
- Data-quality profiling, cleaning, and transformation logic.
- A combined processed master dataset with a city identifier.
- A dimensional model for analytical reporting.
- DuckDB SQL queries and explanatory visualisations.
- Exploratory KMeans listing segmentation and segment profiling.
- A Streamlit dashboard that consumes curated outputs.
- Documentation covering decisions, limitations, deferred work, and AI usage.

2.3 Prioritisation Rationale

The assessment includes a broad range of possible deliverables. The work was therefore prioritised around high-value foundations: data engineering quality, reproducibility, transparent analysis, a functioning dashboard, and honest documentation. Advanced operational features such as retry handling, metadata-driven orchestration, cloud deployment, formal statistical testing, and forecasting were documented as deferred rather than claimed as completed.

Scope boundary

The solution is an analytical prototype. It is not a production booking system, a confirmed-occupancy forecasting engine, a profitability prediction model, or an investment recommendation tool.

3

Source Data and Cross-City Harmonisation

Understanding source differences before combining data is essential for responsible analysis.

3.1 Source Datasets

Three source domains were used for each city: listings, reviews, and calendar data. Listings contain property, host, location, room-type, pricing, and availability attributes. Reviews provide guest-review event records that can be aggregated to listing level. Calendar data contains day-level availability indicators and is the largest source component.

City	Listings	Review records / summary availability	Calendar scale
London	96,871	2,097,996 raw review records	35M+ daily calendar records
New York	35,036	Review-level data available for aggregation	12M+ daily calendar records
Combined	131,907	Listing-level review metrics created	Aggregated to listing-level measures

3.2 Cross-City Harmonisation Strategy

Both cities were mapped into a common analytical structure. A `city` field was added during master-dataset construction so that city provenance remains visible in every downstream output. Common analytical fields include listing identifier, location, room type, property type, cleaned source price, review metrics, calendar availability metrics, and derived enrichment measures.

3.3 Important Source Differences

- London and New York source prices are not currency-normalized. Cross-city price values are descriptive only.
- London calendar price was not usable as a consistent source for calendar-level revenue analysis; this prevented unsupported revenue or seasonal price claims.
- New York contained substantial source-data gaps in host-date and bed-related variables, limiting comparable host-tenure and price-per-bedroom coverage.
- Calendar unavailable dates do not prove that a listing was booked.

4

Data Profiling and Quality Assessment

Profiling established the quality rules used by the pipeline and analysis.

4.1 Profiling Approach

The profiling workflow inspected schema, data types, record counts, null values, duplicate risk, price distributions, and relationships between listings, reviews, and calendar records. This process established which variables were safe to use directly, which required transformation, and which should be treated with explicit caveats.

4.2 Key Data-Quality Findings

Area	Observed Finding	Treatment
Price	Price values were stored as formatted strings and contained missing values.	Converted to numeric `price_clean`; missing values preserved as null.
Reviews	Listings without review records produce missing review aggregation values.	Interpretation distinguishes zero/no review history from a data-processing failure.
Calendar	Availability is stored at day level and requires aggregation.	Converted to listing-level available/unavailable day counts and rates.
Beds and bathrooms	Coverage varies materially by city.	Derived fields only calculated where values are valid; invalid denominators avoided.
Host tenure	New York host-start information is not consistently available.	Host-tenure metrics retained only where source data permits.

4.3 Profiling Decision

The project does not erase limitations through broad imputation. Instead, missingness and source differences are retained as part of the analysis context. This is particularly important for a cross-city dataset because an apparently clean field can represent different collection methods or coverage in each city.

Quality principle

No feature was treated as universally comparable merely because it shared a column name across cities. Each feature was profiled before it was used in analysis, modelling, or dashboard logic.

5

Data Cleaning and Standardisation

Transformation rules were implemented both transparently and modularly.

5.1 Listing Cleaning Rules

- Convert source price strings by removing currency symbols and separators, then parse numeric values into `price_clean`.
- Parse available date columns with error coercion to prevent malformed source values from failing the pipeline.
- Flag high source-price values using a percentile-based outlier threshold rather than deleting them from the master dataset.
- Derive host tenure where both listing scrape date and host-start date are available.
- Protect ratio calculations such as price per bedroom from zero or invalid denominators.

5.2 Price-Outlier Strategy

Price outliers are retained and flagged in the processed dataset because deleting them would weaken traceability. A stricter outlier exclusion is used only in the KMeans clustering subset, where a small number of extreme values can dominate a distance-based algorithm and produce an artificial segment.

5.3 Cleaning Outcome

The cleaned outputs create a consistent numerical basis for enrichment, SQL analytics, visualisation, and exploratory segmentation while preserving raw-data limitations for later interpretation. The pipeline remains reusable because cleaning functions are maintained in source modules rather than only in notebook cells.

6

Data Enrichment and Master Dataset Construction

Row-level source records were transformed into an analytical listing-level grain.

6.1 Review Aggregation

Review records were aggregated by listing identifier to produce listing-level metrics including total review count and first/last review dates. Aggregating reviews prevents a one-to-many relationship from duplicating listing records in the final master dataset.

6.2 Calendar Aggregation

Calendar records were aggregated by listing identifier into total observed days, available days, unavailable days, availability rate, and occupancy/unavailability proxy. This step is essential because the raw calendar sources contain tens of millions of daily records and are not suitable for direct dashboard-level use.

6.3 Master Dataset Construction

Cleaned listings were joined with review and calendar summaries through validated left joins. The resulting master dataset preserves one row per listing while adding city context and derived metrics. The combined result contains 131,907 listings across London and New York and serves as the central analytical dataset for SQL, visualisation, segmentation, and the dashboard.

Derived Field	Definition	Interpretation Caveat
availability_rate	Available calendar days / total observed calendar days	Represents availability status, not demand alone.
occupancy_rate	Unavailable calendar days / total observed calendar days	Proxy only; unavailable dates are not confirmed bookings.
total_reviews	Count of aggregated review records per listing	Historical engagement measure, not a quality score.
price_per_bedroom	Cleaned price divided by bedrooms where valid	Only populated where bedroom values are reliable.
host_tenure_years	Difference between scrape date and host start date where available	Coverage differs between cities.

7

Dimensional Data Model

A star-style structure supports reusable reporting and SQL analysis.

7.1 Model Components

Table	Role	Examples of Key Fields
dim_city	City dimension	city_id, city
dim_neighbourhood	Location dimension	neighbourhood_id, city_id, neighbourhood_cleansed
dim_listing	Listing attribute dimension	listing_id, room_type, property_type
fact_listing_performance	Analytical fact table	price_clean, availability_rate, occupancy_rate, total_reviews, review_frequency

7.2 Analytical Grain

The analytical fact table is designed at the listing level. This makes it possible to combine dimensions such as city, neighbourhood, room type, and property type with metrics such as price, availability proxy, total reviews, review frequency, price per bedroom, and host tenure.

7.3 Modelling Controls

- City was retained as a formal analytical dimension so that cross-city analysis does not lose source context.
- Neighbourhood identifiers were linked to the relevant city, avoiding ambiguity when location names could overlap.
- The project checked for cross-city listing-ID collisions before treating listing identifiers as a unified business key.
- The model is exported as curated CSV tables for transparent local use with DuckDB and Python.

7.4 Star-Schema Relationship Justification

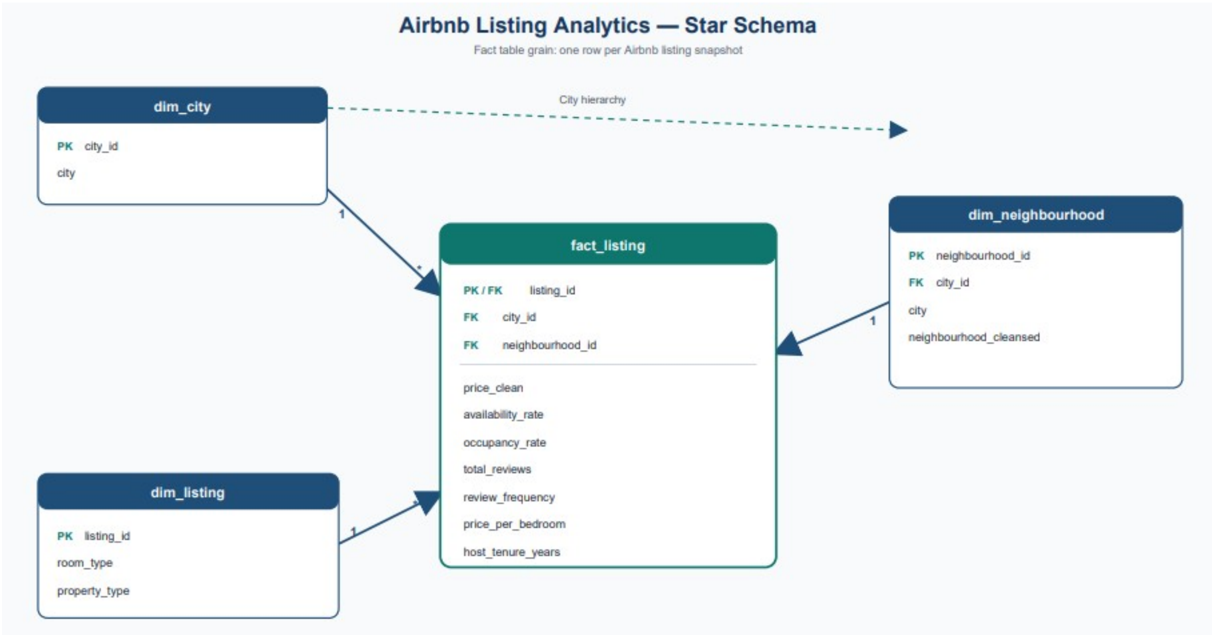

Figure 1. Airbnb listing analytics star-style schema. The diagram uses the short label fact_listing; the exported curated table is named fact_listing_performance.

- dim_city.city_id is the primary key for the city dimension and has a one-to-many relationship with fact_listing_performance through city_id.
- dim_neighbourhood.neighbourhood_id is the primary key for the neighbourhood dimension and has a one-to-many relationship with fact_listing_performance through neighbourhood_id.
- dim_neighbourhood also contains city_id. This creates a small snowflake extension that prevents neighbourhood names from losing their city context.
- dim_listing.listing_id maps one-to-one to the fact table at the current listing-snapshot grain. It stores descriptive attributes such as room type and property type.
- Raw reviews and raw calendar data were one-to-many child datasets. They were aggregated to listing level before joining to the fact table, preventing duplicated listing rows and double-counted measures.

Model design trade-off

The model is intentionally star-style rather than a fully normalized transactional schema. It prioritises clear analytical joins and reproducible reporting while preserving city-scoped neighbourhood identity.

8

Repeatable Pipeline Design

Modular source code replaces one-off notebook-only processing.

8.1 Pipeline Modules

Stage	Module	Responsibility
Ingestion	src/ingestion/load_data.py	Loads city-level listings, reviews, and calendar inputs.
Validation	src/profiling/validate_data.py	Checks required datasets, identifiers, and non-empty inputs.
Cleaning	src/cleaning/clean_listings.py	Cleans price fields, parses dates, flags outliers, derives host tenure where possible.
Review aggregation	src/transformation/review_aggregation.py	Summarises review activity at listing level.
Calendar aggregation	src/transformation/calendar_aggregation.py	Summarises day-level calendar records into listing metrics.
Master builder	src/transformation/build_master.py	Joins cleaned listings with review and calendar summaries.
Orchestration	src/pipeline.py	Runs multi-city processing and writes processed outputs.

8.2 Execution Flow

Raw Listings + Reviews + Calendar Files

↓

Validation and Cleaning Modules

↓

Review and Calendar Aggregations

↓

Combined Processed Master Dataset

8.3 Design Boundary

The Streamlit dashboard does not rerun the full pipeline interactively. This prevents a user-interface action from repeatedly processing very large calendar files and maintains a clear architecture: source modules produce curated outputs; the dashboard reads and presents them.

9

SQL Analytics and Exploratory Data Analysis

DuckDB queries and visualisations expose patterns while preserving data caveats.

9.1 DuckDB Analytical Queries

DuckDB was used to query curated dimensional outputs locally. The main analyses covered city-level listing supply, average source price, calendar-unavailability proxy, price patterns by room type, premium neighbourhoods within each city, and highly reviewed listings.

9.2 Key SQL Findings

Topic	Finding	Interpretation
Listing supply	London contains 96,871 listings; New York contains 35,036.	London has the larger listing inventory in the analysed extracts.
Calendar proxy	London average proxy ≈ 0.603; New York ≈ 0.570.	London listings were marked unavailable on a slightly larger share of observed dates.
Source prices	London average source price ≈ 229.92; New York ≈ 255.00.	These numbers must not be directly compared as a currency-normalized market ranking.
Premium neighbourhoods	Premium location queries were ranked independently inside each city.	Within-city ranking avoids misleading cross-currency comparisons.
Room types	Hotel-room and entire-home categories show broader and higher price distributions.	Room-type differences are stronger visual evidence than raw city price ranking.

Analytics rule

Cross-city comparisons are strongest for record counts and calendar-derived proxy rates. Source price comparisons are limited to within-city patterns because currencies are not normalized.

9.3

Price Distribution Analysis

Price charts are displayed using the 99th-percentile cap for readability; the underlying data remains unchanged.

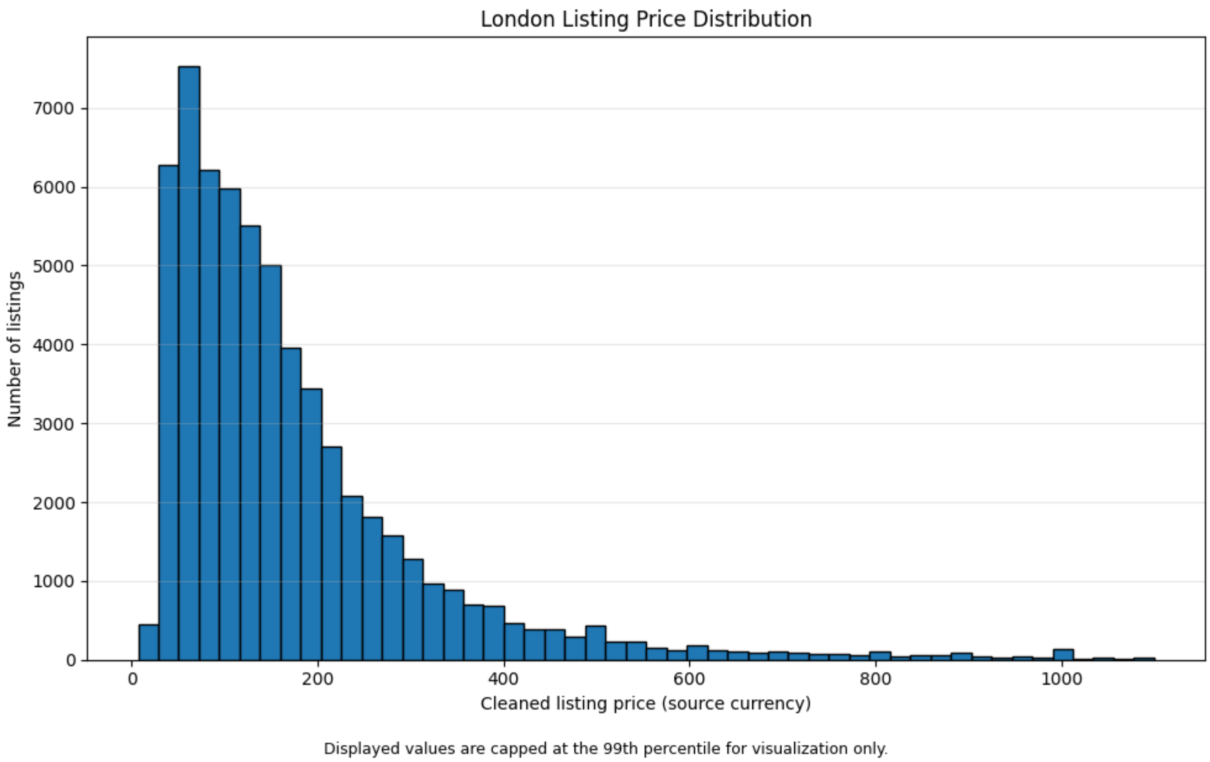

Figure 2. London listing price distribution. Values displayed up to the 99th percentile for visual readability.

The London distribution is right-skewed: most listings occur at lower-to-mid source-price levels, while a smaller high-price tail extends toward the display cap. This supports the decision to flag outliers for traceability and to avoid allowing a small number of extreme values to dominate exploratory analysis.

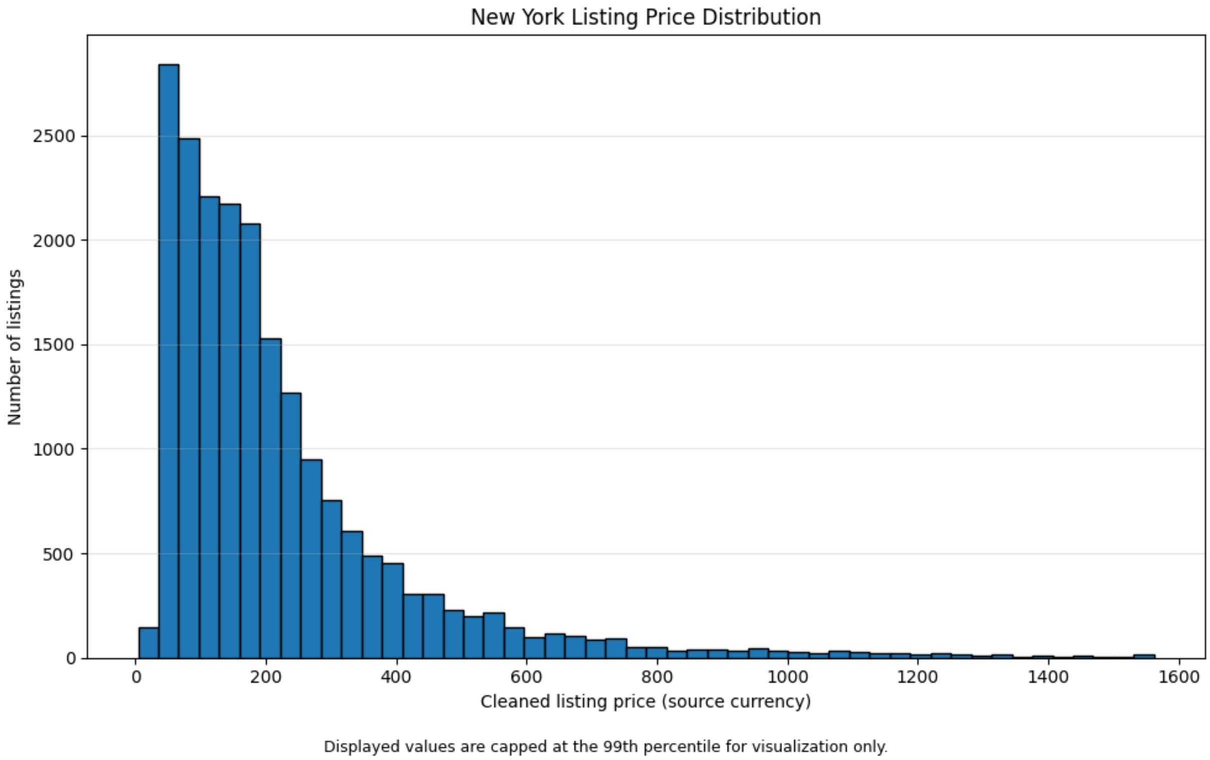

Figure 3. New York listing price distribution. Values displayed up to the 99th percentile for visual readability.

The New York distribution is also right-skewed with a long upper tail. The chart should be interpreted as a within-city distribution, not as a conversion-adjusted comparison against London.

9.4

Room-Type Price Analysis

Room type is a useful analytical dimension because it offers a consistent within-city comparison.

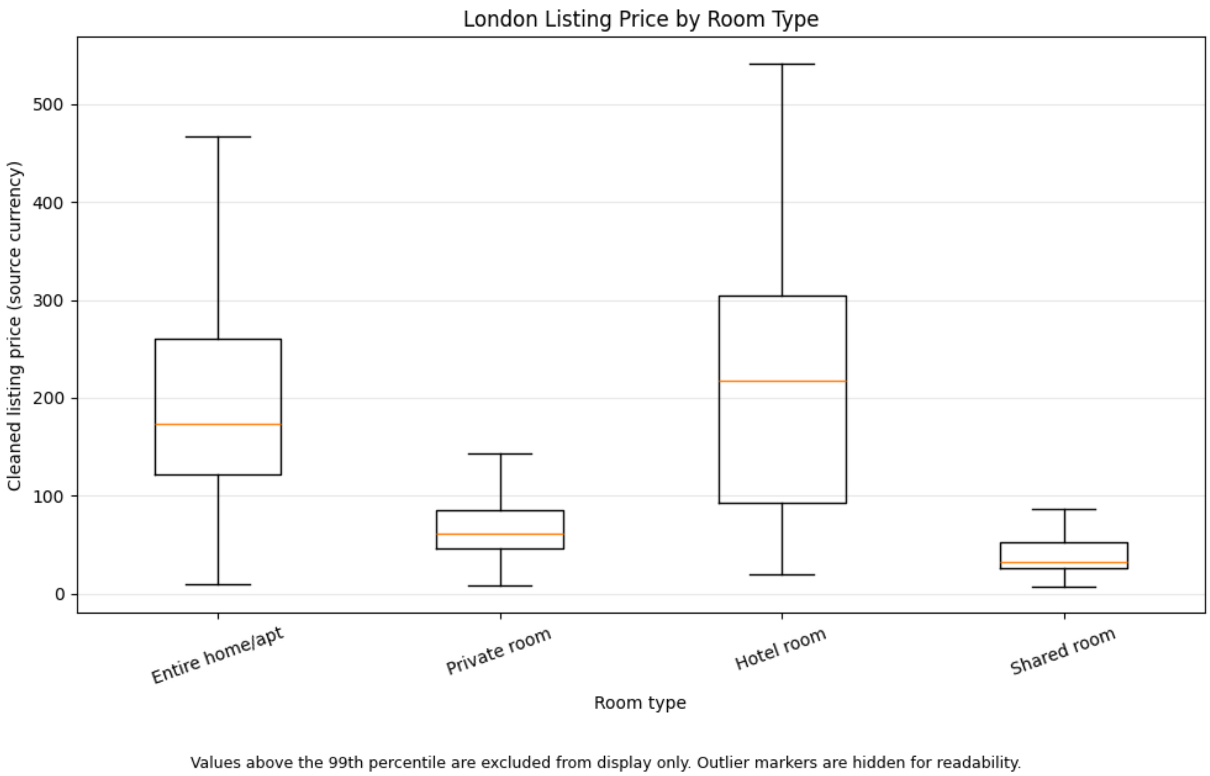

Figure 4. London listing price by room type. Display excludes values above the 99th percentile and hides outlier markers for readability.

London hotel rooms and entire homes/apartments show higher central price ranges than private and shared rooms. The wider spread for hotel rooms and entire homes/apartments indicates greater pricing heterogeneity within those categories.

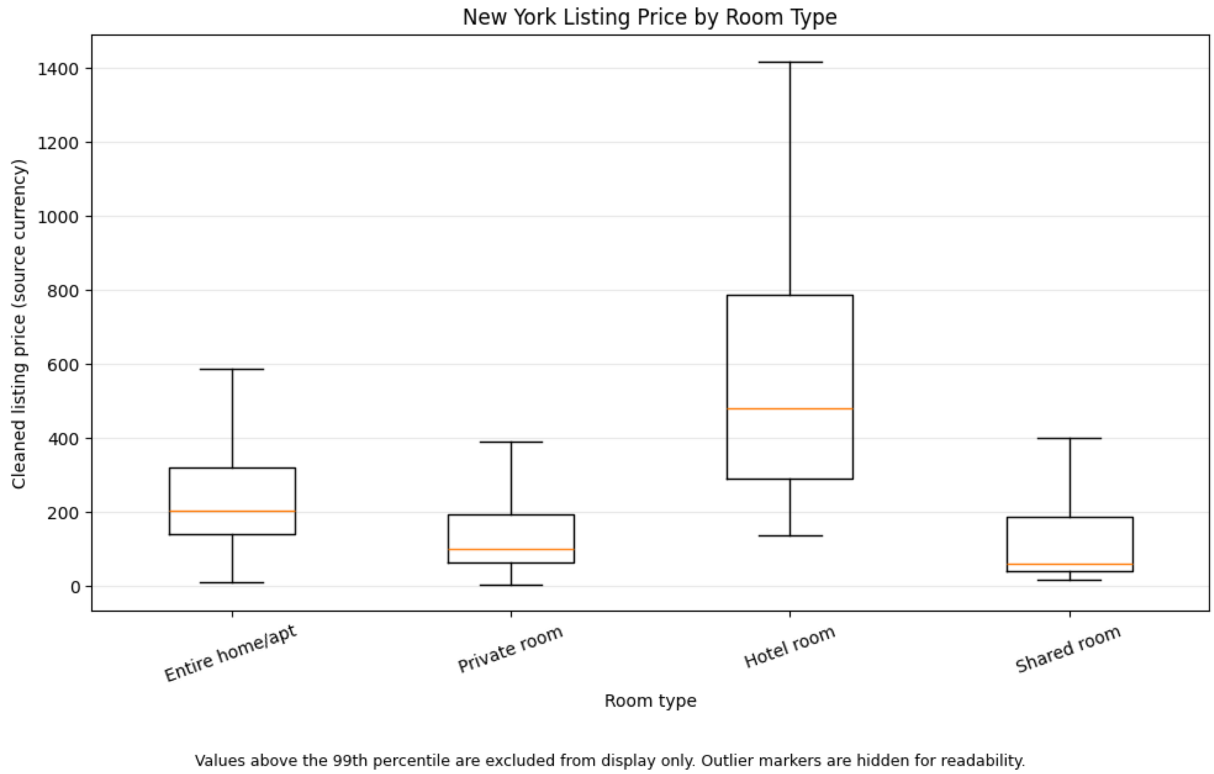

Figure 5. New York listing price by room type. Display excludes values above the 99th percentile and hides outlier markers for readability.

New York exhibits the same broad pattern: hotel rooms and entire homes/apartments occupy a higher price range, while private and shared rooms are generally lower. This room-type segmentation is more suitable for comparison than direct city-level price ranking.

9.5

Statistical Thinking: Room-Type Price Comparison

A city-specific non-parametric hypothesis test complements the descriptive visual analysis.

9.5.1 Business Question and Hypotheses

Business question: within each city, are Entire home/apt listings priced higher than Private room listings? The analysis was run separately for London and New York so that local source currencies were never mixed in one statistical test.

- Null hypothesis (H0): the Entire home/apt source-price distribution is not higher than the Private room source-price distribution within the city.
- Alternative hypothesis (H1): the Entire home/apt source-price distribution is higher than the Private room source-price distribution within the city.

9.5.2 Test Choice and Assumptions

A one-sided Mann-Whitney U test was selected because the observed price distributions were right-skewed and contained high-value extremes. The test does not require a normal distribution. Each row represents one listing, price is a continuous/ordered measure, and listing observations were treated as independent for this exploratory analysis. A 99th-percentile cap was applied only to the test subset to reduce the influence of extreme source-price values; the main master dataset was not changed.

9.5.3 Results

City	Entire home/apt n	Private room n	Median entire	Median private	Median difference	Mann-Whitney U	Adjusted p-value	Rank-biserial r	Effect
London	41,770	19,315	173.00	61.00	112.00	733,069,377.5	< 0.001	0.8173	Large
New York	10,857	9,108	203.50	100.21	103.30	73,489,015.0	< 0.001	0.4863	Medium

The raw numerical output underflowed to 0.0 because both p-values are extremely small. They are therefore reported responsibly as $p < 0.001$. A Bonferroni adjustment was applied for the two city-level tests, and both results remain statistically significant.

9.5.4 Business Interpretation

The evidence supports rejecting H0 for both cities. Entire home/apt listings have substantially higher source-price distributions than Private rooms within both London and New York. The London effect is large (rank-biserial correlation 0.8173), while the New York effect is medium (0.4863). This means room type is a meaningful within-city pricing dimension and should be considered when benchmarking listings, setting pricing strategy, or building future market segments. The median differences must not be compared across cities as a currency-normalized financial result.

Statistical conclusion

Room type is not merely visually different: the within-city difference between Entire home/apt and Private room source-price distributions is statistically significant after adjustment in both cities, with medium-to-large practical effect sizes.

10

Exploratory KMeans Segmentation

The machine-learning component groups similar listing behaviours into interpretable analytical segments.

10.1 Purpose

KMeans clustering was used to create an exploratory segmentation of listings. The objective was not to predict bookings, profitability, or investment return. Instead, clustering was used to identify patterns in calendar availability/unavailability behaviour, review activity, and source-price characteristics.

10.2 Preparation

- Records with incomplete clustering features were excluded from the clustering subset only.
- A 99.9th-percentile source-price cap was applied to the clustering subset to reduce distortion from extreme values.
- Features were standardised before KMeans because price, review counts, and rates operate on different numerical scales.
- The original master dataset was not overwritten by the modelling subset.

10.3 Model Selection

The final selected configuration used four clusters. The refined clustering evaluation produced a silhouette score of approximately 0.505, indicating a useful level of separation for an exploratory operational segmentation. The result was interpreted alongside cluster sizes and business readability rather than treated as a universal model-quality guarantee.

10.4 Elbow and Silhouette Evidence

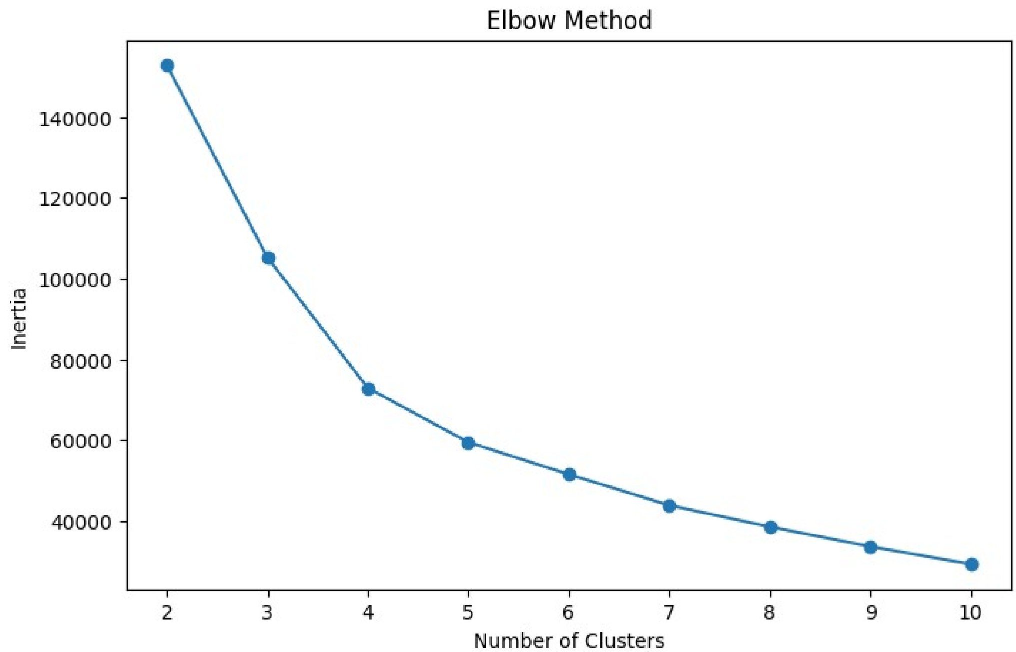

Figure 6. Elbow-method evaluation. Inertia falls sharply up to approximately four clusters and then reduces more gradually.

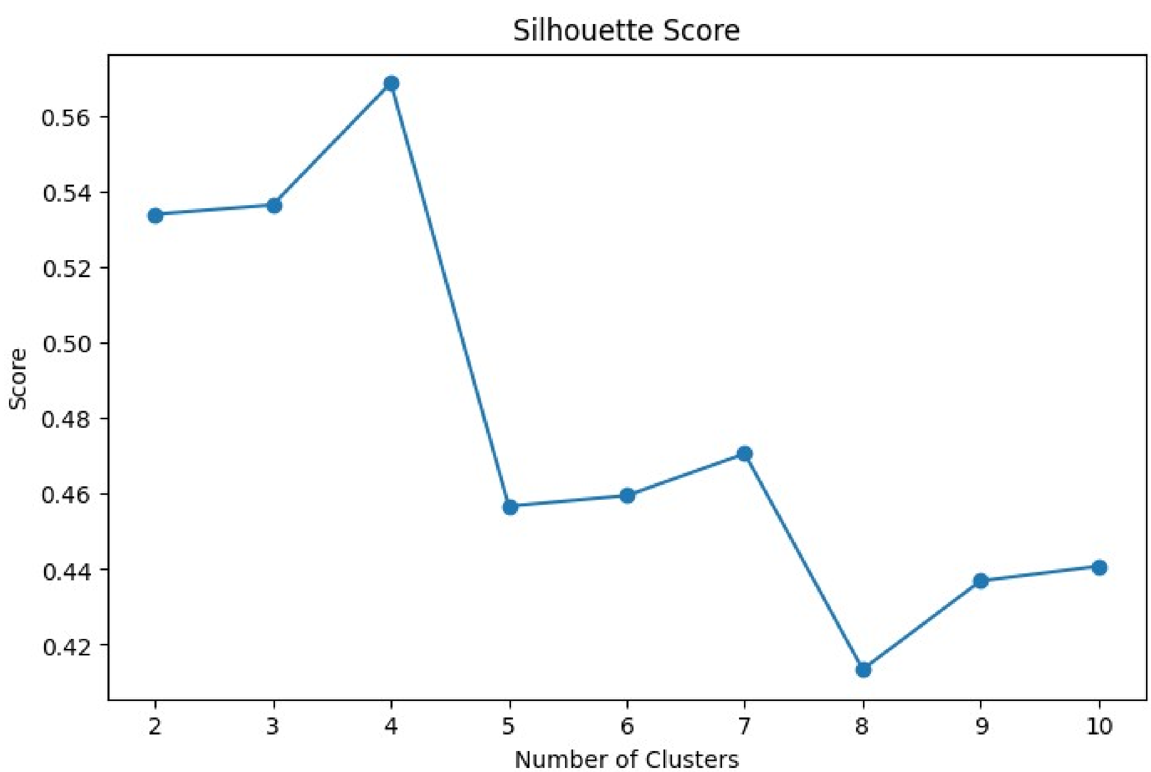

Figure 7. Initial silhouette-score evaluation. k = 4 achieved the highest score at approximately 0.569 before the final extreme-price refinement.

The initial screening evaluation selected k = 4 because it produced the highest silhouette score, approximately 0.569. After applying the 99.9th-percentile price cap to remove the influence of a very small extreme-price group from the modelling subset, the final four-cluster model retained a silhouette score of approximately 0.505. This provides useful exploratory separation but is not presented as a production-grade classification metric.

10.5 Final Segment Definitions

Segment	Primary Pattern	Responsible Interpretation
High Demand	Highest average calendar unavailability proxy.	May indicate consistently limited availability, but not confirmed bookings.
Established Popular	High historical review activity.	Signals established guest engagement, not necessarily current quality or demand.
Premium Luxury	Higher source-price profile.	Premium interpretation is exploratory and not currency-normalized across cities.
Low Performance	Lowest calendar unavailability proxy.	May indicate an opportunity to review pricing, visibility, content, or availability practices.

11

Guided Analytics Assistant and Streamlit Dashboard

The UI demonstrates how curated engineering and analytics outputs can be made accessible to users.

11.1 Dashboard Design

The Streamlit application was designed as a presentation and exploration layer. It reads the processed master dataset and curated segmentation outputs, ensuring that dashboard interactions do not duplicate pipeline logic or attempt to process raw calendar data in the browser session.

11.2 Dashboard Components

- Market Overview: listing counts, city coverage, segmentation coverage, city-level availability proxy, and descriptive source-price summary.
- Guided Analytics: validated business questions with outputs calculated from curated data.
- Segment Explorer: profile metrics, explanatory notes, and listing examples for the selected exploratory segment.
- Pipeline Status: transparent evidence that the dashboard depends on the repeatable pipeline and curated outputs.

Controlled assistant design

The Guided Analytics Assistant is intentionally constrained to validated analytical questions. It is not presented as a free-text LLM chatbot. This keeps outputs reproducible, traceable, and aligned with actual data logic.

11.3 Dashboard Evidence: Market Overview

The first dashboard view communicates the high-level analytical scope and the necessary interpretation caveats.

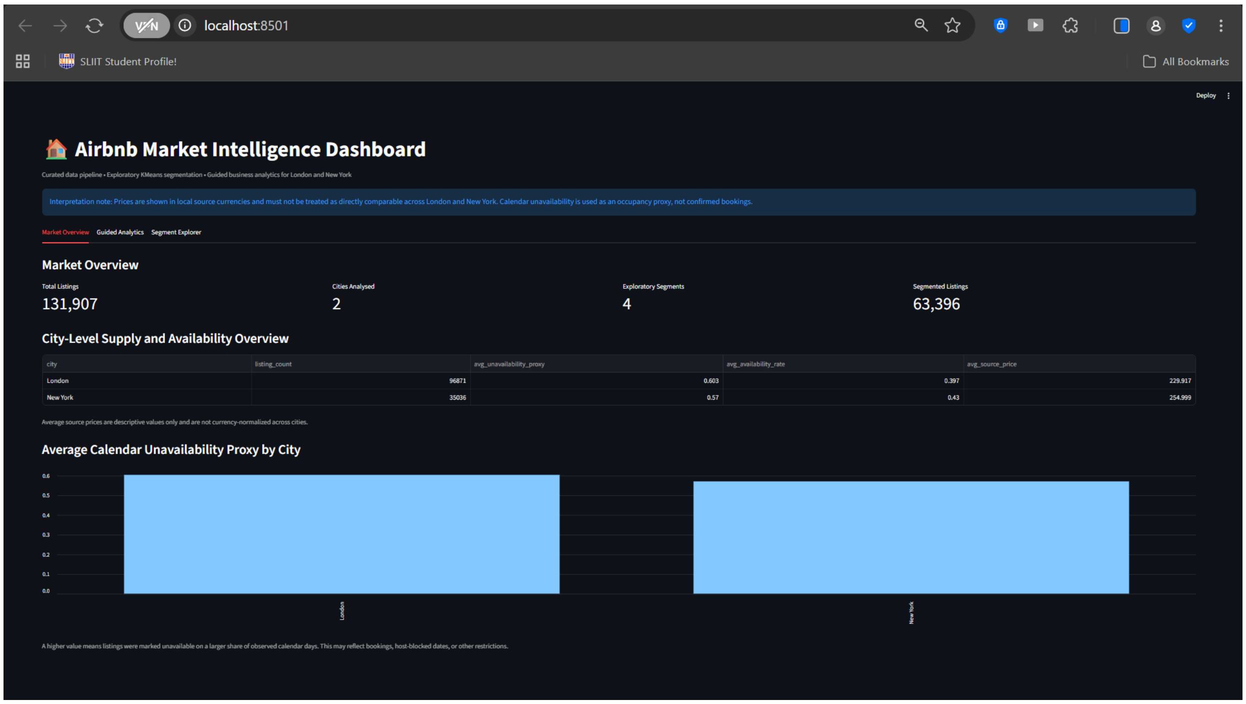

Figure 8. Streamlit Market Overview showing listing totals, city coverage, segment coverage, and average calendar-unavailability proxy.

The Market Overview makes the dashboard boundaries visible before users explore results. It states that prices are shown in local source currencies and must not be used as a direct cross-city price comparison. It also describes the calendar metric as an occupancy/unavailability proxy rather than confirmed occupancy.

11.4

Dashboard Evidence: Guided Analytics

Validated questions return transparent curated-data answers instead of ungrounded narratives.

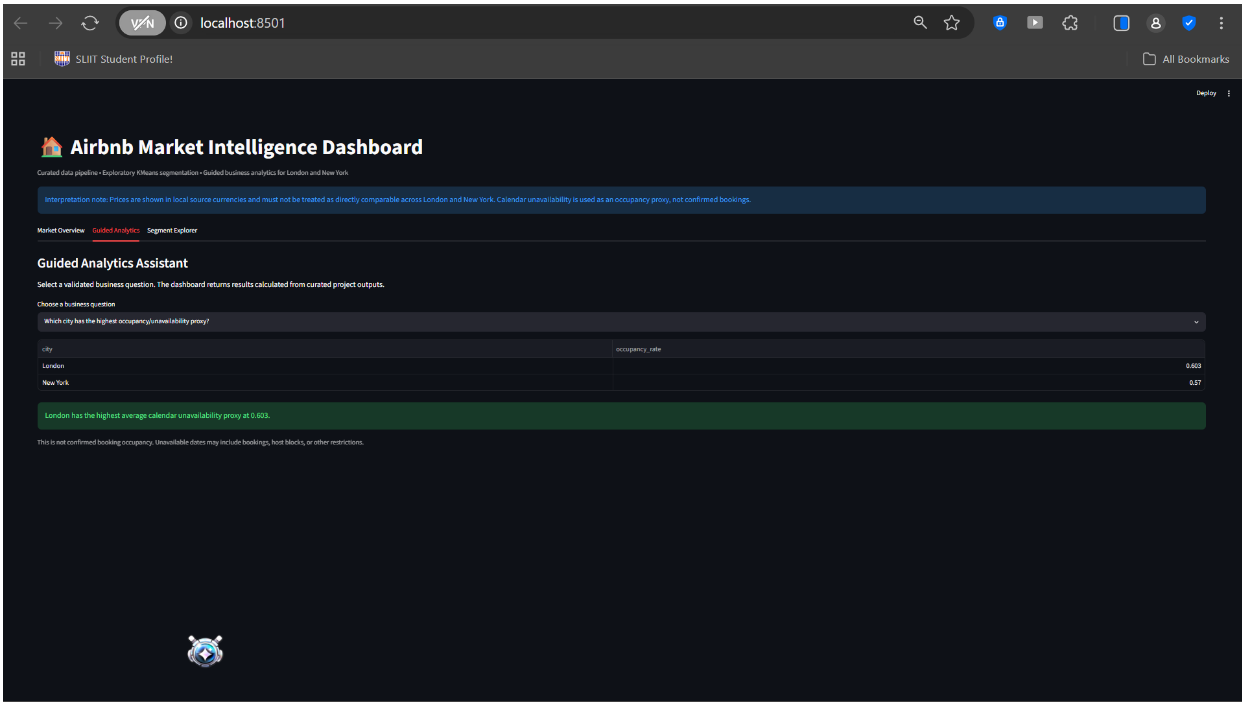

Figure 9. Guided Analytics Assistant showing the city-level calendar-unavailability proxy comparison.

This view demonstrates a controlled analytical question: which city has the highest average calendar-unavailability proxy? The result identifies London at approximately 0.603 and New York at approximately 0.570, while the interface repeats the caveat that unavailable dates can include host blocks or other restrictions.

11.5

Dashboard Evidence: Segment Explorer

Segment-level outputs combine model results with plain-language interpretation.

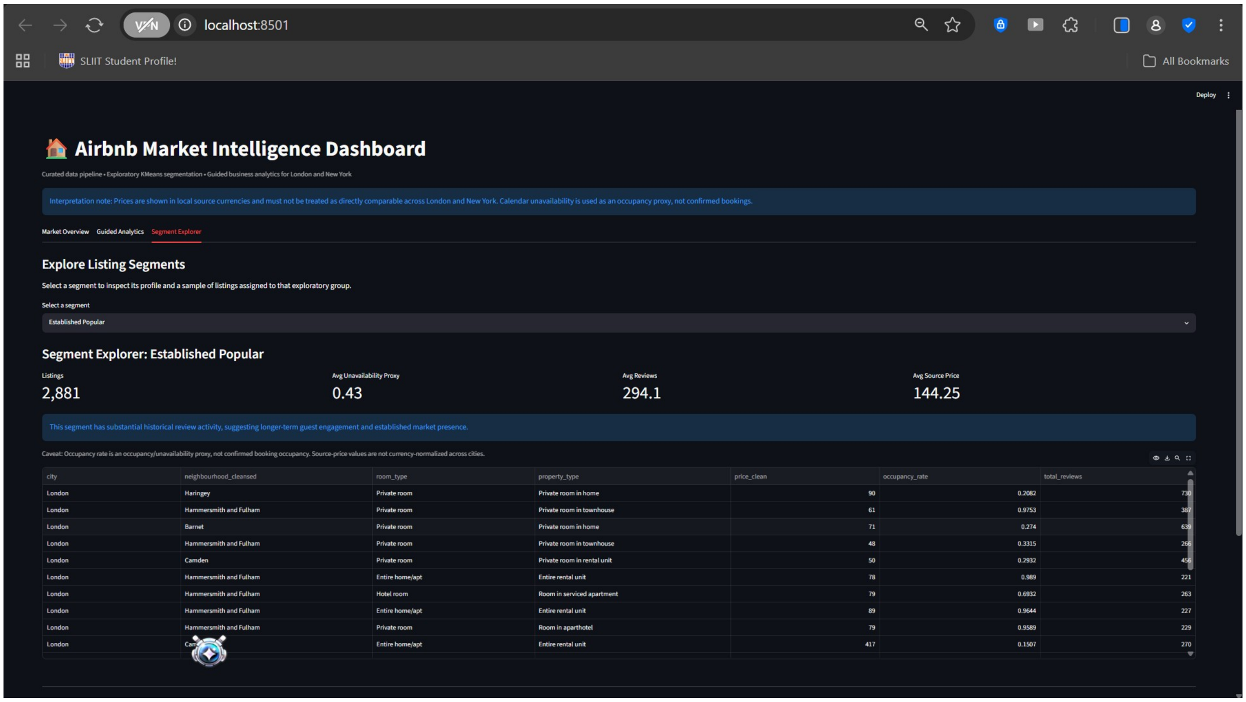

Figure 10. Segment Explorer showing the Established Popular segment, its summary metrics, explanation, and sample listings.

The Segment Explorer converts KMeans output into a human-readable analytical tool. The shown Established Popular segment contains 2,881 listings in the example view and exhibits high average historical review activity. The interface retains caveats about calendar proxy interpretation and cross-city source prices.

12

Cross-City Findings and Business Recommendations

Recommendations are positioned as exploratory actions, not definitive market guarantees.

12.1 Findings

- London has a substantially larger listing supply in the analysed source extracts than New York.
- London has a slightly higher mean calendar-unavailability proxy than New York in the processed master dataset.
- Both cities show right-skewed source-price distributions, making median and distribution-based views more informative than a simple global average.
- Room type materially changes price distribution within each city; entire homes/apartments and hotel rooms show higher and wider price ranges.
- Review activity provides a separate behavioural signal that helps distinguish established listings from lower-engagement listings.

12.2 Suggested Analytical Actions

Stakeholder Need	Suggested Action	Evidence Source
Review low-engagement inventory	Inspect Low Performance listings by neighbourhood, room type, source price, and listing content.	Segment Explorer and curated segmentation output.
Study established supply	Inspect Established Popular listings to identify common location and property-type patterns.	Review aggregation and KMeans segment profile.
Compare neighbourhood positioning	Use city-specific median-price ranking with a minimum listing-count threshold.	DuckDB premium-neighbourhood query.
Maintain operating discipline	Use availability proxy alongside caveats; do not equate unavailable dates with confirmed bookings.	Calendar aggregation design and dashboard notes.

12.3 Recommendation Boundary

The findings are suitable for exploratory market analysis, operational review, and discussion. They are not sufficient to recommend a property investment, guarantee demand, or estimate financial return without currency normalization, verified booking status, cost data, and time-series validation.

13

Limitations, Deferred Work, and Future Roadmap

Transparent boundaries improve the credibility of the completed solution.

13.1 Key Limitations

- Calendar unavailability is a proxy rather than confirmed occupancy or booking data.
- Source prices are not currency-normalized, so cross-city price figures are descriptive only.
- Coverage differs across cities for host tenure, beds, bathrooms, and related derived measures.
- KMeans segments are exploratory analyst interpretations, not natural or ground-truth classes.
- The dashboard presents curated analytical results; it is not an online operational data-processing system.

13.2 Deferred Work

Area	Deferred Capability	Reason / Next Step
Statistics	Additional hypothesis tests and confidence intervals	One focused Mann-Whitney U test was completed; broader statistical validation remains future work.
Pipeline operations	Structured logging, retries, incremental loads, orchestration	Requires a production operating environment and additional engineering effort.
Financial comparison	Historical exchange-rate normalization	Needed before responsible cross-city price benchmarking.
ML validation	Alternative clustering algorithms and stability testing	Appropriate future enhancement beyond an exploratory baseline.
Deployment	Cloud storage, scheduled jobs, CI/CD	The current assessment solution is a reproducible local implementation.

13.3 Future Roadmap

The strongest next step is to build from the existing modular foundation rather than replace it. A production roadmap would add configuration-driven city ingestion, structured pipeline logs, data-quality checks, incremental processing, currency normalization, scheduled orchestration, and deeper statistical or forecasting components.

14

Conclusion and Reflection

The completed work demonstrates a full data journey from raw records to usable analytical products.

The project delivers a cohesive Airbnb market intelligence platform rather than a disconnected set of notebook exercises. Raw listings, reviews, and calendar files are transformed through reusable modules into a combined processed master dataset. The curated outputs then support SQL analysis, visual exploration, exploratory segmentation, and a user-facing Streamlit dashboard.

A key lesson from the work is that technical completeness is not enough without interpretive discipline. Calendar status cannot be labelled as confirmed bookings, and London/New York source prices cannot be treated as directly comparable without normalization. These limitations were incorporated into the design, the dashboard wording, the decision log, and the report rather than hidden.

The final product therefore demonstrates both engineering execution and analytical accountability: modular pipeline logic, reproducible outputs, clear data modelling, controlled user interaction, transparent limitations, and practical pathways for future enhancement.

Final delivery statement

Raw Airbnb data is transformed through a modular pipeline into processed and curated outputs that power transparent SQL analytics, exploratory KMeans segments, and a Streamlit market intelligence dashboard.

A

Appendix A - Deliverable Inventory

Files included in the project repository

Deliverable	Location	Purpose
Dataset familiarisation notebook	notebooks/01_dataset_familiarization.ipynb	Introduces sources, schema, relationships, and assumptions.
Data profiling notebook	notebooks/02_data_profiling.ipynb	Quality checks, missingness, duplicates, and price profiling.
Data cleaning notebook	notebooks/03_data_cleaning.ipynb	Cleaning logic, parsing, outlier treatment, and validation.
Data enrichment notebook	notebooks/04_data_enrichment.ipynb	Review/calendar aggregation and master dataset construction.
Data modelling notebook	notebooks/05_data_modeling.ipynb	Dimensional model and fact/dimension exports.
SQL analytics notebook	notebooks/06_sql_analytics.ipynb	DuckDB analysis and report-ready charts.
Pipeline design notebook	notebooks/07_pipeline_design.ipynb	Modular pipeline execution evidence.
AI analyst notebook	notebooks/08_ai_analyst_assistant.ipynb	KMeans segmentation and guided analytics logic.
Streamlit app	app.py	Presentation layer over processed and curated outputs.
Decision log	docs/decision_log.md	Decisions, assumptions, and rationale.
Completion summary	docs/completed_vs_deferred.md	Completed and deferred work.
AI disclosure	docs/ai_usage_disclosure.md	Transparent AI-assisted development disclosure.

B

Appendix B - Reproducibility Guide

Suggested execution sequence for a local environment

B.1 Environment Setup

- Create and activate a Python virtual environment.
- Install project dependencies from requirements.txt.
- Ensure raw London and New York files are placed under the expected data/raw directory structure.

B.2 Recommended Execution Order

Step	Action	Expected Output
1	Run notebooks 01-06 in sequence.	Profiling, cleaning, enrichment, model outputs, SQL analysis, and figures.
2	Run the modular pipeline through src/pipeline.py or Notebook 07.	data/processed/ pipeline_master_dataset.csv
3	Run Notebook 08.	Curated segmentation files.
4	Launch Streamlit with `streamlit run app.py`.	Local Market Intelligence Dashboard.
5	Review docs/ documentation files.	Decision log, scope disclosure, and AI usage disclosure.

B.3 Key Output Files

- data/processed/pipeline_master_dataset.csv
- data/curated/dim_city.csv
- data/curated/dim_neighbourhood.csv
- data/curated/dim_listing.csv
- data/curated/fact_listing_performance.csv
- data/curated/ai_segmented_listings.csv
- data/curated/ai_segment_profile.csv

C

Appendix C - AI Collaboration and Validation Record

AI assistance was disclosed, constrained, reviewed, and critically evaluated throughout the project.

C.1 AI Tools Used

Tool / Model	Use During Assignment	Model Version Disclosure
ChatGPT	Technical mentoring, debugging, documentation structure, report drafting, analytical review.	GPT-5.5 Thinking
VS Code Local Agent	Local coding assistance and implementation support within the development environment.	Model version not separately recorded
Claude	Code and documentation assistance, analytical discussion, and review support.	Claude Sonnet 4.6

C.2 AI-Assisted Sections

- Python pipeline and notebook code review, debugging, and explanation.
- Notebook Markdown, README, decision log, completion summary, and report drafting.
- Review of dashboard wording and removal of unsupported claims about bookings, investment performance, and cross-city currency comparisons.
- Analytical-method discussion for KMeans, source-price treatment, data modelling, and the Mann-Whitney U test.

C.3 Representative Prompts Used

Prompt 1 - Senior technical review

"You are my Senior Data Engineering Mentor and Technical Reviewer for the Expernetic Data Engineer Intern Assignment. My objective is not merely to finish it, but to produce a technically sound, well-documented, reproducible data-engineering solution that stands out."

Prompt 2 - Requirement alignment

"Review the assessment brief against my completed notebooks, pipeline modules, dashboard, and documentation. Identify missing requirements, unsupported claims, and the highest-value fixes before submission."

Prompt 3 - Analytical quality control

"Check this analysis for cross-city currency issues, proxy-metric interpretation risks, outlier treatment, and business claims that are not supported by the data."

Prompt 4 - Report drafting constraints

"Create a professional report from the actual project outputs. Include limitations and deferred work honestly; do not invent results, statistics, or production capabilities."

C.4 Output Validation and Testing

Validation Activity	How It Was Performed
Notebook execution	The author ran the familiarisation, profiling, cleaning, enrichment, modelling, SQL, pipeline, and AI notebooks locally and checked outputs.
Data-output validation	Shapes, required columns, duplicate checks, missingness patterns, outlier flags, and city-specific assumptions were reviewed.
Pipeline validation	Reusable modules were executed through the pipeline workflow for London, New York, and the combined master dataset.
Dashboard validation	The Streamlit dashboard was launched locally and checked through Market Overview, Guided Analytics, Segment Explorer, and Pipeline Status views.
Visual validation	Charts, diagrams, and report pages were rendered and visually inspected for clear labels, caveats, and consistent presentation.

C.5 Meaningful Author Modifications

- Changed dashboard and report wording from confirmed bookings or investment recommendations to calendar-unavailability proxy and exploratory analytical insight.
- Replaced misleading cross-city raw-price rankings with city-specific neighbourhood price rankings and explicit currency caveats.
- Kept price outliers flagged in the core pipeline but limited extreme-price exclusion to the KMeans subset and focused statistical subset only.
- Removed the uploaded-city dashboard prototype from the final user interface because it duplicated pipeline logic and was less reliable for the final demonstration.
- Added a city-specific Mann-Whitney U test with Bonferroni adjustment and rank-biserial effect size rather than relying only on visual price comparison.

C.6 Critical Assessment and Rejected Suggestions

A late alternative KMeans feature rewrite was tested but rejected. Its silhouette scores were approximately 0.313, materially weaker than the established exploratory segmentation result after outlier refinement (approximately 0.505). The author reverted the proposed rewrite rather than accepting it blindly. This decision preserved a more interpretable and stable final model.

The author also rejected unsupported language that implied verified booking activity, direct currency-normalized price comparisons, or guaranteed investment opportunities. These changes were reflected in the final dashboard, documentation, and report.

AI-use conclusion

AI was treated as a structured co-pilot, not an authority. Prompts specified the role, context, constraints, deliverable format, and validation expectations; the author executed, reviewed, modified, or rejected outputs based on evidence.

D

Appendix D - Statistical and Modelling Evidence

Supporting evidence for inferential analysis, data modelling, and KMeans selection.

D.1 Statistical Output Summary

The Mann-Whitney U analysis compared Entire home/apt and Private room source-price distributions separately within London and New York. Both city-level tests were significant after Bonferroni adjustment ($p < 0.001$), with a large effect in London and a medium effect in New York. Full values are reported in Section 9.5 and saved in data/curated/statistical_test_room_type_price.csv.

D.2 Dimensional Model Nomenclature

The visual schema uses fact_listing as a compact diagram label. The repository export uses the full filename fact_listing_performance.csv. They refer to the same listing-grain analytical fact concept described in Section 7.

D.3 KMeans Reproducibility Notes

- KMeans was run with n_clusters=4, random_state=42, and n_init=10.
- Features were standardised prior to distance-based clustering.
- The final modelling subset excluded only values above the 99.9th percentile of price_clean to reduce extreme-price distortion.
- The original master dataset remained unchanged and segment labels were assigned after cluster profiling for stakeholder readability.

D.4 Reproducible Evidence Files

- notebooks/05_data_modeling.ipynb - dimensional model construction.
- notebooks/06_sql_analytics.ipynb - SQL, visual analysis, and statistical test.
- notebooks/08_ai_analyst_assistant.ipynb - KMeans selection, refinement, segments, and guided analytics.
- data/curated/statistical_test_room_type_price.csv - statistical result table.
- docs/decision_log.md and docs/ai_usage_disclosure.md - decision and AI-use documentation.